

# VIVAAN THADANI

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## SUMMARY OF QUALIFICATIONS

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- Mechanical engineering student with hands-on experience in CAD modelling, mechanical drawings, rapid prototyping, design validation, and structural analysis.
- Created and modified 3D CAD models using SolidWorks and AutoCAD to support prototype development, assembly planning, and fabrication workflows.
- Supported R&D-style engineering projects through physical testing, wind tunnel testing, iterative design, technical documentation, and performance improvement.
- Demonstrated ability to solve open-ended engineering problems from concept development through fabrication, testing, and deployment in real-world operating environments.
- Prepared engineering reports, technical data, workflow procedures, parts lists, and preliminary mechanical BOM documentation while collaborating across mechanical and electrical teams.

## SKILLS

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- **Engineering Software:** SolidWorks (CSWA), ANSYS, Fusion 360, AutoCAD, Bambu Studio
- **Core Engineering Skills:** CAD Modelling, Mechanical Drawings, GD&T, Design Validation, Structural Analysis
- **Programming and Data Tools:** MATLAB, Python, C, Microsoft Excel

## EDUCATION

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### Toronto Metropolitan University

Sep 2024 – April 2029 (Expected)

*Bachelor of Engineering, Mechanical Engineering, cGPA: 3.83/4.33*

*Dean's List: 2024, 2025*

- **Relevant Coursework:** Manufacturing, Engineering Design, Statics, Mechanics of Materials, Dynamics, Thermodynamics, Circuits

## EXPERIENCE + PROJECTS

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### Mechanical Engineering Intern – Toronto Water

May 2026 – Aug 2026

- Designed and created a solar powered wiper for a live feed sewer camera using Fusion 360, Arduino and 3D printing for the sewer environment, resulting in a maintenance cost reduction of \$6000 annually per camera site.
- Designed and fabricated custom sensor caps using CAD software, 3D printing, and CNC machining to ensure secure and reliable installation of monitoring equipment in confined storm water and sewage environments.
- Developed and validated a sensor inventory application through edge-case testing and issue tracking; created user guides and workflow documentation and trained 40+ operators on asset tracking and system use.

### Structures Project Lead – TMU Hyperloop

Oct 2025 – Present

- Lead R&D of a PCM-assisted water-glycol cooling system for LIMs by validating materials through literature review and coordinating piping/enclosure design, BOM development, and weekly design reviews.
- Performed structural FEA in ANSYS Mechanical on LIM mounts and 2024 aluminum brackets by calculating loads, defining materials, meshes, and boundary conditions, and documenting stress/deformation results.
- Coordinated mechanical and electrical interfaces across vehicle systems to improve subsystem compatibility and support cross-functional integration.

### Gear-Driven Autonomous Wind-Up Car – MEC322 Design Project

Jan 2026 – Apr 2026

- Collaborated in a cross-functional team of seven to deliver a fully functional mechanical prototype from concept to competition, managing design integration, manufacturing constraints, and iterative testing within a structured engineering design process.
- Performed engineering calculations for gear geometry, torque transmission, drive force, and travel distance to validate a 3:1 cycloid gear reduction within strict material and dimensional constraints.
- Developed a complete SolidWorks CAD drawing package with assembly and detail drawings and motion simulation, then fabricated and iteratively tested a laser-cut acrylic prototype.